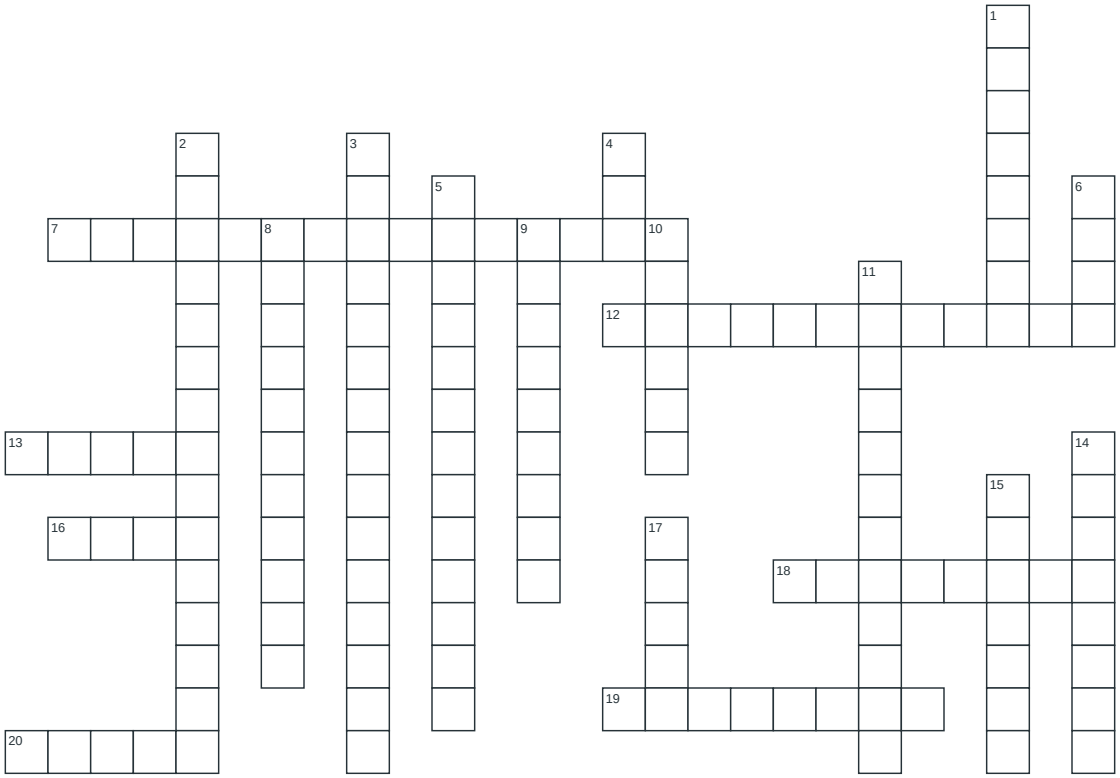
Motion Speed and Acceleration Crossword

Hard | 20 clues | Student Worksheet

Name: _____ Date: _____

Directions: Use the clues to complete the grid. Omit spaces, punctuation, and hyphens.



ACROSS

7. Forces between interacting objects are equal in magnitude and opposite in direction.
12. The directed change from initial to final position
13. Definition of the SI unit of energy and work
16. Measure of the amount of matter in an object.
18. Exerts a force that changes the passenger's motion
19. Force that opposes relative motion between surfaces in contact.
20. Quantity that is a scalar

DOWN

1. The vector sum of all forces acting on an object
2. SI-derived unit commonly used to measure speed or velocity
3. Energy associated with position in a gravitational field

DOWN (continued)

4. Relationship that expresses Newton's second law for constant mass
5. Energy of motion; for a fixed mass it varies with the square of speed.
6. State in which position does not change relative to a chosen reference point
8. Force that is a surface's push perpendicular to that surface
9. Orientation of motion, needed to describe a velocity vector
10. The gravitational force acting on an object
11. Change in velocity per unit time
14. The total length of a traveled path
15. Resistance to changes in motion
17. The rate of energy transfer or work

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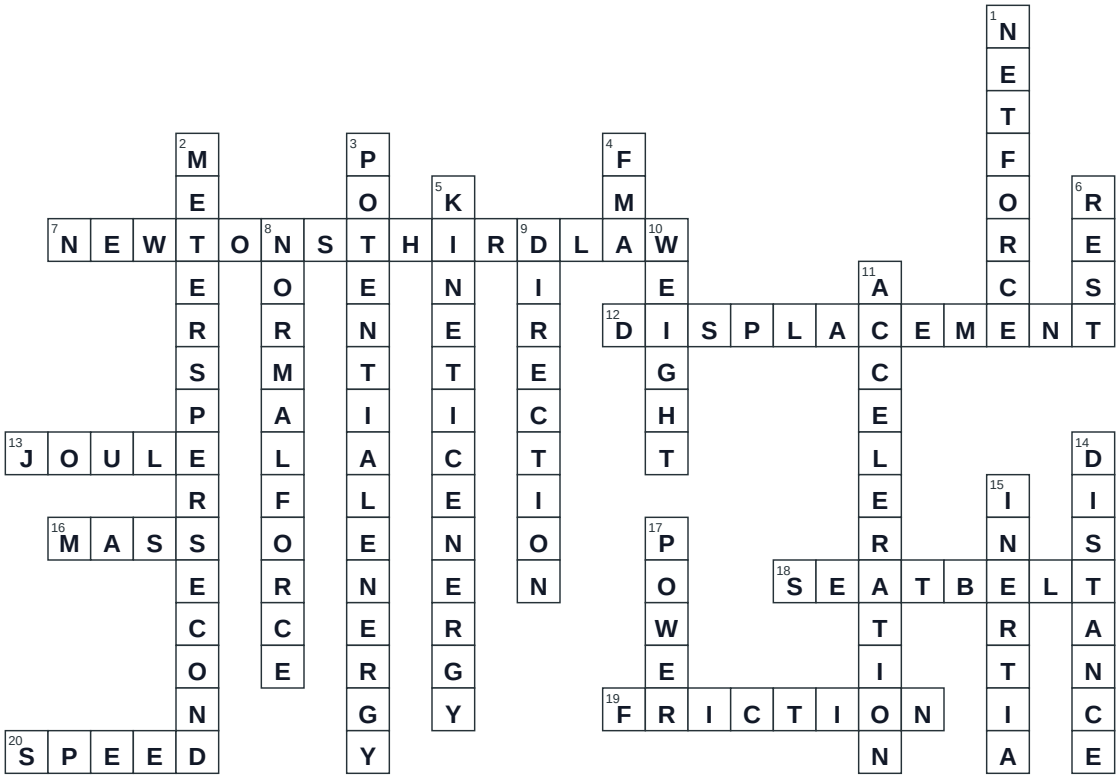
Hard | 20 clues | Teacher Answer Key

TEACHER ANSWER KEY

Completed grid and numbered answers. Spaces and punctuation are omitted in the grid.



ACROSS

7. Newton’s third law
12. Displacement
13. Joule
16. Mass
18. Seat Belt
19. Friction
20. Speed

DOWN

1. Net Force
2. Meters Per Second

DOWN (continued)

3. Potential Energy
4. F = ma
5. Kinetic energy
6. Rest
8. Normal Force
9. Direction
10. Weight
11. Acceleration
14. Distance
15. Inertia
17. Power

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